MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 06
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The sum of the exponents of prime factors in the prime factorisation of 196 is: [1]

(a) 1 (b) 2 (c) 4 (d) 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 4

Prime factorising 196:

Formula Used: $196 = 4 \times 49 = 2^2 \times 7^2$

Sum of prime exponents $= 2 + 2 = \boxed{4}$. [1M]

Q2. The sum of the exponents of prime factors in the prime factorisation of 144 is: [1]

(a) 4 (b) 5 (c) 6 (d) 7

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 6

Prime factorising 144:

Formula Used: $144 = 16 \times 9 = 2^4 \times 3^2$

Sum of prime exponents $= 4 + 2 = \boxed{6}$. [1M]

Q3. If two positive integers m and n are expressed as $m = p^2q^3$ and $n = p^3q$, where p and q are primes, then $\text{HCF}(m, n) \times \text{LCM}(m, n)$ is: [1]

(a) p^5q^4 (b) p^6q^3 (c) p^5q^3 (d) p^6q^4

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) p^5q^4

We know the product identity for two positive integers:

Formula Used: $\text{HCF}(m, n) \times \text{LCM}(m, n) = m \times n$

$m \times n = (p^2q^3)(p^3q^1) = p^{2+3}q^{3+1} = \boxed{p^5q^4}$. [1M]

Q4. If p_1 and p_2 are two odd prime numbers such that $p_1 > p_2$, then $p_1^2 - p_2^2$ is always: [1]

(a) an odd integer (b) an even integer
(c) an odd prime (d) a fractional number

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) an even integer

Since p_1 and p_2 are both odd integers, their squares p_1^2 and p_2^2 are both odd. We know:

Formula Used: $p_1^2 - p_2^2 = (p_1 - p_2)(p_1 + p_2)$, $\quad$ (Odd) $\pm$ (Odd) = Even

Both $(p_1 - p_2)$ and $(p_1 + p_2)$ are even (in fact divisible by 4), e.g., $5^2 - 3^2 = 16$. Hence $p_1^2 - p_2^2$ is $\boxed{\text{an even integer}}$. [1M]

Q5. Assertion (A): $2 + \sqrt{3}$ is an irrational number.
Reason (R): The sum of a rational number and an irrational number is always an irrational number. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

We know rational-plus-irrational arithmetic closure:

Formula Used: $r \in \mathbb{Q},\ x \notin \mathbb{Q} \implies r + x \notin \mathbb{Q}$

If $2 + \sqrt{3} \in \mathbb{Q}$, then $\sqrt{3} = (2 + \sqrt{3}) - 2 \in \mathbb{Q}$, contradicting irrationality of $\sqrt{3}$. Hence both (A) and (R) are true and (R) explains (A). [1M]

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. Find the prime factorisation of the number 32760 and write it in exponential form. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

We know the Fundamental Theorem of Arithmetic guarantees a unique prime-power decomposition:

Formula Used: $N = p_1^{a_1} \cdot p_2^{a_2} \cdots p_k^{a_k} \quad (p_1 < p_2 < \cdots < p_k \text{ primes})$

Dividing step by step by ascending prime factors:

$$32760 \xrightarrow{\div 2} 16380 \xrightarrow{\div 2} 8190 \xrightarrow{\div 2} 4095 \xrightarrow{\div 3} 1365 \xrightarrow{\div 3} 455 \xrightarrow{\div 5} 91 \xrightarrow{\div 7} 13. \qquad [1M]$$

Collecting identical prime factors in exponential form:

$$32760 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 13 = \boxed{2^3 \times 3^2 \times 5 \times 7 \times 13}. \qquad [1M]$$

SECTION C: Short Answer Type — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Find the largest positive integer that will divide 398, 436, and 542 leaving remainders 7, 11, and 15 respectively. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Subtracting the respective remainders makes each number exactly divisible by the required integer d:

$$398 - 7 = 391, \qquad 436 - 11 = 425, \qquad 542 - 15 = 527. \qquad [1M]$$

We know the required largest divisor is:

Formula Used: Required largest divisor $d = \text{HCF}(391,\ 425,\ 527)$

Prime factorising the reduced numbers:

$$391 = 17 \times 23, \qquad 425 = 5^2 \times 17, \qquad 527 = 17 \times 31. \qquad [1M]$$

Only prime 17 divides all three, so $\text{HCF}(391, 425, 527) = \boxed{17}$. [1M]

Q8. Find the least number which when divided by $20, 25, 35$, and 40 leaves remainders $14, 19, 29$, and 34 respectively. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Checking the difference between each divisor and its corresponding remainder:

$$20 - 14 = 6, \qquad 25 - 19 = 6, \qquad 35 - 29 = 6, \qquad 40 - 34 = 6. \qquad [1M]$$

Because a constant difference $d = 6$ holds across all divisors, we apply:

Formula Used: Least number $= \text{LCM}(20, 25, 35, 40) - d$

Prime factorising: $20 = 2^2 \times 5$, $25 = 5^2$, $35 = 5 \times 7$, $40 = 2^3 \times 5$.

$$\text{LCM}(20, 25, 35, 40) = 2^3 \times 5^2 \times 7 = 8 \times 25 \times 7 = 1400. \qquad [1M]$$

Subtracting common difference 6: Required number $= 1400 - 6 = \boxed{1394}$. **[1M]**

Q9. Find the smallest number which leaves remainder 8 and 12 when divided by 28 and 32 respectively. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Observing the constant difference between divisor and remainder:

$$28 - 8 = 20 \qquad \text{and} \qquad 32 - 12 = 20. \qquad [1M]$$

We know that for constant divisor–remainder difference $d = 20$:

Formula Used: Smallest number $N = \text{LCM}(28, 32) - 20$

Prime factorising 28 and 32:

$$28 = 2^2 \times 7, \qquad 32 = 2^5 \implies \text{LCM}(28, 32) = 2^5 \times 7 = 32 \times 7 = 224. \qquad [1M]$$

Subtracting 20: $N = 224 - 20 = \boxed{204}$ (Check: $204 = 28 \times 7 + 8$ and $204 = 32 \times 6 + 12$). **[1M]**

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Q10. CASE STUDY 06: REPUBLIC DAY ARMY CONTINGENT PARADE MARCHING [4 Marks]

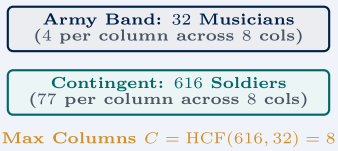
During the annual Republic Day parade rehearsals at Kartavya Path, an army contingent of 616 soldiers is scheduled to march behind an army band of 32 musicians. The parade drill commander mandates that the two groups must march in the exact same number of columns, with each column possessing a uniform alignment and discipline.

(i) Write the prime factorisations of 616 and 32. [1]

(ii) Find the maximum number of columns in which the band and contingent can march. [2]

(iii) How many contingent soldiers and band members will march in each column respectively? [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Parade Formation: Every column contains an exact integer count of musicians from 32 and soldiers from 616. Maximizing columns selects their greatest common divisor, $\text{HCF}(616, 32) = 8$.

We know the HCF uniform-column formula:

Formula Used: Max equal columns = $\text{HCF}(616, 32)$

(i) Prime factorisations:

$$616 = 8 \times 77 = \boxed{2^3 \times 7 \times 11}, \qquad 32 = \boxed{2^5}. \qquad [1M]$$

(ii) Maximum columns: Taking smallest exponent of common prime 2:

$$\text{HCF}(616, 32) = 2^3 = \boxed{8 \text{ columns}}. \qquad [2M]$$

(iii) Personnel in each column:

$$\text{Soldiers per column} = \frac{616}{8} = \boxed{77}, \qquad \text{Band members per column} = \frac{32}{8} = \boxed{4}. \qquad [1M]$$

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